

The Human Cell Atlas from a cell census to a unified foundation model

<https://doi.org/10.1038/s41586-024-08338-4>

Received: 16 September 2024

Accepted: 1 November 2024

Published online: 20 November 2024

 Check for updates

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With the convergence of notable advances in molecular and spatial profiling methods and new computational approaches taking advantage of artificial intelligence and machine learning, the construction of cell atlases is progressing from data collection to atlas integration and beyond. Here, we explore five ways in which cell atlases, including the Human Cell Atlas, are already revealing valuable biological insights, and how they are poised to provide even greater benefits in the coming years. In particular, we discuss cell atlases as censuses of cells; as 3D maps of cells in the body, across modalities and scales; as maps connecting genotype causes to phenotype effects; as 4D maps of development; and, ultimately, as foundation models of biology unifying all of these aspects and helping to transform medicine.

Understanding how our cells function in health and disease is essential both for basic human biology and for medical advances in diagnosis and treatment. While genetic studies have mapped more than 100,000 disease-associated variants in the human genome, we do not know in which cells the majority of these variants are active, and which function they perform. Without this knowledge, we cannot fully understand biology, study more powerful models of disease, deploy better diagnostics and develop more effective therapies. The challenge is vast: the human body has an estimated 37.2 trillion cells, spanning an incredible diversity of types and states, and each of those cells is also impacted by our inherent diversity—in ancestry, geography, age, gender and life experiences—and changes continuously over time.

The Human Cell Atlas (HCA) initiative, a consortium of several thousand scientists launched in 2016, aims to tackle this challenge by creating comprehensive reference maps of all human cells, as a basis for both understanding human health and diagnosing, monitoring and treating disease^{1,2}. Atlases are broadly defined as collections of maps. In geography (cartography), these typically cover various geographies, and could provide multiple views of the same geography (topography, roads, political boundaries, climate, demographics). In mathematics (differential geometry), an atlas is a collection of charts that, intuitively, describe various parts of a manifold, together allowing for a coherent transition between different local descriptions. The HCA borrows from both concepts—analogue to a geographical atlas, cellular maps can cover different organs (geographies) and provide different views of these cells, including which cells constitute a tissue or organ, which dynamic states they can assume, how they transition between states and where the cells are located, and, when viewed as a collection, these views can give a broader perspective of the human body, beyond a single region. Like a manifold, we can view each cell as mapped to a point in a continuous, high-dimensional space based on its molecular or other features, with individual cell types or states represented as local charts. These charts cover parts of the manifold and provide a detailed, local view of cellular diversity, and algorithms can serve to transition smoothly between different cellular states (further

discussion of the cell atlas concept is available in the HCA White Paper³ and foundational perspective⁴). Generating such an atlas became possible using new laboratory and computational technologies, including large-scale single-cell and spatial genomics, and the substantial progress in artificial intelligence and machine learning (AI/ML) and data science. Since its inception, the HCA has become a major driving force at the nexus of biology, technology and medicine. It has been a partner to or a model for multiple efforts by related initiatives, focused on disease biology (for example, the Human Tumour Atlas Network⁵ and the Kidney Precision Medicine Project⁶) and specific organs (for example, the BRAIN initiative⁷ and the Gut Cell Atlas⁸), and multimodal (such as the NIH HubMAP^{9,8}), multiscale (CIFAR MacMillan Multi-Scale Human project (<https://cifar.ca/research-programs/cifar-macmillan-multiscale-human/>)) and model-organism (for example, The Fly Cell Atlas⁹) atlases.

After initial planning and data-gathering phases, the HCA community is now in an era of integration and assembly of a first draft atlas, focused on the main data collected to date: single-cell-resolution molecular profiles, mostly of RNA. We report many efforts towards data collection, data integration and computational methods development in a large collection of papers published concurrently with this Perspective. Together, these advances show that it is now possible to gather, model, analyse and query ever-larger collections of single-cell and spatial genomic data, opening exciting vistas for the atlas as a foundation map for cell and tissue biology. Here we explore five current and future perspectives of the cell atlas: as a cell census, a 3D map, a temporal developmental map, a genotype-to-phenotype map and a multimodal foundation model for cell biology (Fig. 1). We highlight considerable progress in each of these areas and the remaining challenges, and how these uses of cell atlases are set to impact biomedicine.

The cell atlas as a census

Constructing a comprehensive reference map requires discovering, cataloguing and describing the cell types and states present across all tissues, organs and systems in a systematic, coherent, and

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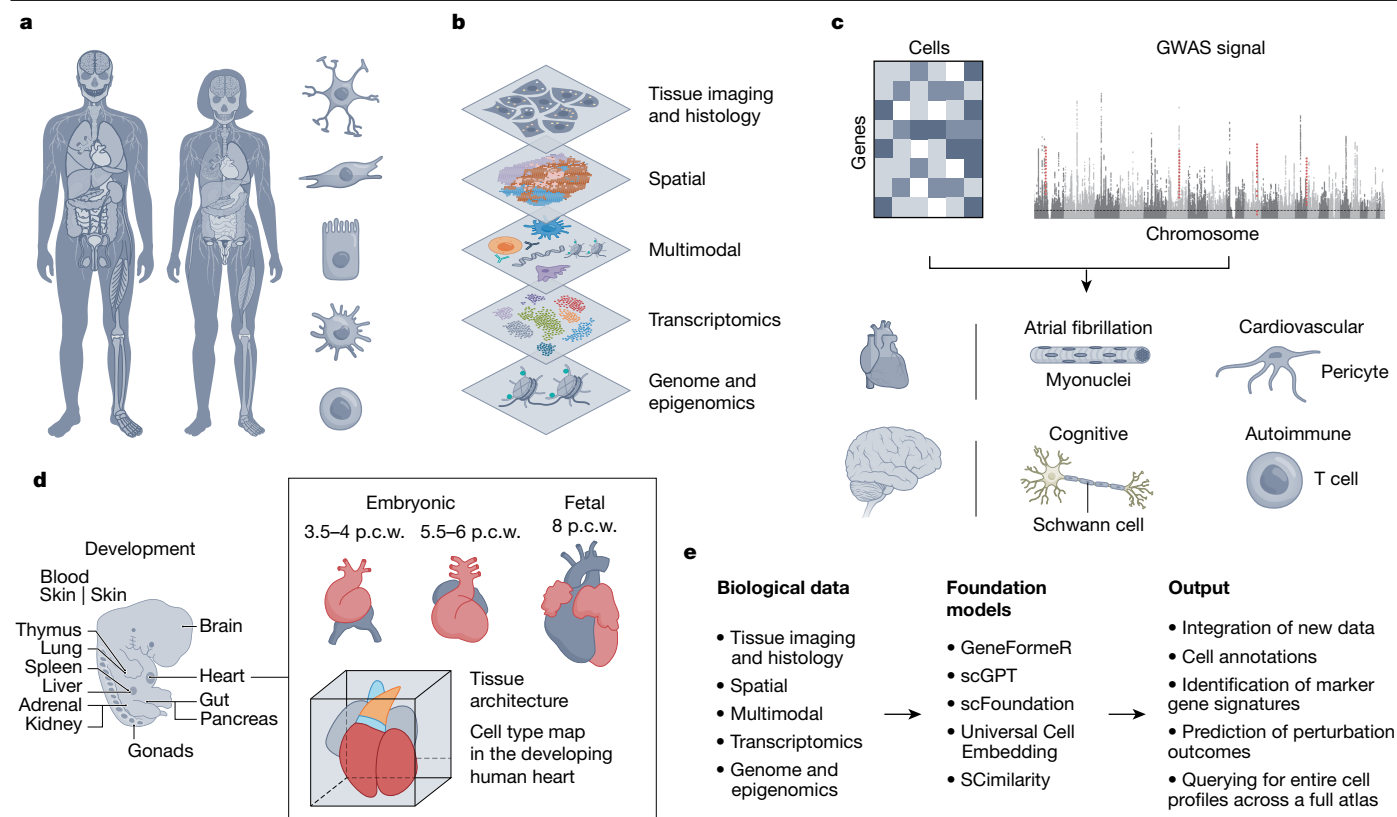


Fig. 1 | Five perspectives on cell atlases. **a–e**, Cell atlases can be censuses of cells within a particular organ or system (**a**); multiscale, multimodal 3D maps of cells and tissues within the body (**b**); maps connecting genotype (that is, from GWAS analyses) to phenotype at the cellular, tissue or whole-body level (**c**);

4D maps of development that provide a framework and resource for comparison to tissue samples from model systems and diseases (**d**); and, by combining these layers together using computational advances in AI/ML, foundational models of biology (**e**). p.c.w., post-conception weeks.

computationally and biologically rigorous way. This requires both large-scale, comprehensive experiments across multiple biological modalities, both molecular and non-molecular (for example, morphological), at the single-cell and spatial level, and the development of accompanying algorithms for data integration, interpretation and analysis, as well as generating testable predictions that fill in the gaps of unobserved features or cells.

The single-cell and single-nucleus molecular profiling data collected to date are now being integrated to initial atlases, consisting of annotated tissue and organ reference data objects^{10–15}, that will form the first draft HCA v.1. Analogous to the assembly of the ‘golden path’ for contigs into whole chromosomes in the Human Genome Project¹⁶, this iterative effort is increasingly providing the broad biomedical community a reliable and standardized resource for research, a reference for querying and investigation, and an early ‘ground truth’ to train better algorithms. To ensure that the atlas is open and empowers users, all atlases are available on the HCA Data Portal (<https://data.humancellatlas.org/>) and CELLxGENE Discover¹⁷ portals, which provide pre-computed and highly accessible representations and associated querying tools for a broad range of users. We will iteratively evolve these integrated atlases into a complete and comprehensive census of human cell types and states.

The first four draft HCA atlases—the Human Lung Cell Atlas v.1¹⁰, whole-brain and cortical atlases from the NIH BRAIN Initiative¹⁸ (<https://data.humancellatlas.org/hca-bio-networks/nervous-system>) and the Human Retina Cell Atlas v.1 from the HCA Eye Biological Network¹⁹—have already proven to be powerful and impactful references, as have similar atlases for non-human species^{9,20,21}. Each has led to immediate insights. For example, the lung atlas identified rare cell types, variation between healthy individuals and similarities across

different lung diseases (for example, COVID-19, lung cancer and fibrosis). These draft atlases have enabled researchers to compare disease datasets to the healthy state by identifying similar cell types to a cell type of interest, or drilling down into gene programs and molecular mechanisms driving a particular cellular, tissue or organismal phenotype^{22–24}. Comparing human and non-human atlases, such as mouse and human brain atlases^{18,21}, has provided the context to better understand gaps in human atlases, the extent that model organisms mirror human biology and gaps in the completeness of atlases. Other censuses, yet to be integrated into draft atlases, are also having an impact. For example, the endometrial census shed light on how the endometrium regenerates without scarring during the menstrual cycle, and suggested that decidualized stromal cells and macrophages are dysregulated in endometriosis.

To serve as a reference census, an atlas must be rigorous, informative and community focused, and must evolve iteratively as new data and methods arise. To achieve a rigorous and community-oriented atlas, each of 18 HCA Biological Networks (<https://www.humancellatlas.org/biological-networks/>), led by biological domain experts, collaborates with a central computational integration team with AI/ML and computational biology expertise. Together, they have developed a general framework for atlas integration, with specific adaptations to the unique features of different biological systems (for example, neurons tend to group into more discrete groups, whereas many immune cells follow continuums). First, the team sets specific targets for each atlas (for example, unified developmental trajectories in a developmental atlas; disease-specific cell states in a blood cell atlas). Integration approaches are assessed on the basis of their ability to achieve these targets using a systematic comparison, guiding choices of methods, quality-control thresholds and feature selection across broadly shared

steps. As integration efforts proceed, the teams tackle additional challenges, including the need to describe cells through the lens of distinct types and shared gene programs; and the need to relate cells across different organ/system atlases while capturing distinct niche-specific characteristics^{25–27}.

With the assembly of each atlas, the community learns more about the performance of different approaches to integration, how to develop better benchmarks and what the challenges are of integrating different tissue atlases, especially those lacking a foundational understanding of the ground truth that an atlas is aiming to represent. Notably, algorithms can help reveal blind spots in evaluation metrics that could lead to favouring integration methods and algorithms that miss important biological signals. Such an approach helped to reveal that evaluation frameworks can be easily ‘fooled’ by a toy method²⁸. Systematically improving benchmarking frameworks is an active area of research in the HCA community^{29,30}.

A crucial component for practical impact is the biological annotation of cell types and states. The first paradigm of atlas annotation combined computational analysis (for example, clustering cell profiles, generating differentially expressed gene sets, scoring pre-existing gene signatures) and human expert knowledge to ascribe a name to a cell subset. As a result, the ‘same’ cells may have been labelled by similar, but not identical, labels, and defined by similar, but not identical, features. There are four key levers to harmonize those: the Cell Ontology³¹, which establishes a consistent set of terms and relationships; the HCA Cell Annotation Platform for community-driven creation, exploration, storage and serving of annotations (<https://celltype.info/>); transfer learning, to algorithmically annotate new cells with existing annotations³²; and the Biological Networks, to engage experts in annotation. In parallel, foundation models trained on annotated HCA data^{33,34} can redefine annotation by encompassing all of these features within and across tissues (see the ‘Atlases as foundation models for biology’ section). The HCA community has now developed a cohort of computational methods to generate references (for example, Azimuth³⁵), integrate them with new data (for example, SCArches³²), annotate them (for example, CellTypist, CellHint³⁶) query them (for example, SCimilarity²²) and more (for example, SEACells³⁷, CellRank³⁸). Ultimately, we envision autonomous agents³⁹, equipped with large language models, to draw on atlas data, the literature and human expertise for enhanced annotation, including of disease atlases^{40,41}.

A successful atlas census must be queryable. For example, given the profile of a cell or a group of similar cells, a user should be able to label (annotate) their type, map them to the atlas and find atlas gene programs active in them. If a user has a heterogeneous sample of cells, they should be able to query them all simultaneously. They should be able to query for deviations and differences between their cells or sample and the reference (in gene profiles and proportions). It should also be possible to query the atlas with one or a set of genes and find cells and programs that express them, on absolute and relative scales. Several tools developed by the community enable some of these queries, whereas others have not yet been addressed. These tools include foundation models such as scGPT³⁴ and GeneFormer³³; cell profile search and query tools such as SCimilarity²²; tools for disease-specific analyses, such as MultiMIL⁴²; and cell type annotation and harmonization methods, such as CellHint³⁶. These are just a few components of the actively growing toolbox available for researchers. However, note that there are still key opportunities for new method development, both for new approaches to existing queries, as well as methods that address entirely new queries, such as multicellular configurations. Some of these new queries can be challenging to define generally. Furthermore, as the community develops ever more AI-based querying methods, these need to be combined with mechanisms to support the result with associated evidence (as is used in retrieval augmented generation), and to assess the quality and uncertainty of the outputs to ward against overinterpretation or overconfidence.

The atlas as a multimodal and multiscale 3D map

A cell census alone is not a comprehensive biological atlas, as it does not capture the inherent multimodal, multiscale, 3D nature of the human body, from intracellular organization to microhistology to gross anatomy⁴³. To understand the function of cells, we need to know how they are organized intrinsically, and where they are located extrinsically, across different measurement modalities at the same and different length scales, both with respect to a common coordinate framework and relative to other cells and biological entities⁴³. Thus, the cell atlas must also be a generalizable, multimodal and multiscale map that relates multiple different types of molecular, cellular and visual features in an abstracted 3D representation of tissues, organs and the whole body (<https://cifar.ca/research-programs/cifar-macmillan-multiscale-human/>).

With the development of multimodal atlases, different modalities open the opportunity to learn different facets of biology, as the information encoded in each can be distinct. In particular, changes in chromatin, RNA and protein operate at different timescales, both in terms of the speed of change, and its persistence, and vary in different ways in different biological systems. For example, in a fully differentiated cell, chromatin is relatively invariant to environmental changes, RNA is a leading indicator (reflecting both the current and future looking actions of a cell, as captured in RNA velocity⁴⁴), protein levels are a lagging indicator⁴⁵, while protein modifications are an even more rapid, transient and leading indicator than RNA. Conversely, in developing systems, chromatin is often a leading indicator, whereas RNA lags behind it, because chromatin changes precede RNA changes in differentiation^{46,47}. In differentiated cells with a great degree of remaining plasticity (for example, T helper cells), chromatin can reflect the capacity for both current and future states, whereas RNA mostly reflects the current cell states. Even within the same cell system information content may vary between processes. For example, the cell cycle is more reflected in RNA than in chromatin⁴⁷. Finally, and crucially, for mechanistic studies that aim to tie levels together and infer causal relations, multimodality is invaluable, even if similar cell state information could have been inferred from fewer modalities.

The rise of spatial profiling laboratory methods, clinical imaging and AI advances now enable learning the atlas as a 3D multimodal, multiscale, generative model. Experimentally, it is now possible to profile cells and molecules in 2D sections^{48,49} and whole-mount 3D^{50–52} at the supracellular⁵³, cellular^{54,55} and subcellular^{56–59} scales. Current data collection efforts are shifting rapidly to spatial measurement in situ as a main focus. For example, a human limb development⁶⁰ dataset combined single-cell and spatial profiling to identify two waves of human muscle development, with anatomical segregation between genes linked to brachydactyly and polysyndactyly. Non-molecular imaging techniques will be essential for construction of the atlas as a 3D map. These include routine clinical imaging (magnetic resonance imaging and computed tomography) from living patients, large-volume imaging that is currently applied ex vivo, such as light scattering (for example, XPCi⁶¹ and HiPCT⁶²) or Raman-based imaging⁶³, and techniques that are applicable to cleared or fixed specimens (such as DISCO⁶⁴). Recent examples include light-sheet imaging of whole embryos with markers of the developing skull and skeleton⁶⁵, and 3D-DISCO in the brain⁶⁶. Such imaging, and related metadata, will help to tie high-resolution measurements to location.

Computationally, modern AI/ML opens the possibility to learn multiscale, multimodal representations, with or without common anchors^{67–69}. Such models learn common latent spaces capturing salient features of multiple modalities and their inter-relationships, and can also generate one modality from another after training^{70–72}. There are now models that relate across cells, histology and anatomy^{69,73}, and the maps learned on one dataset or technique have even generalized to mapping data from other techniques, not seen in the training⁶⁹. For example, HCA studies have shown how to train models on

histology (haematoxylin and eosin) and single-cell profiling (single-cell RNA-sequencing) data of tumours, and then generate single-cell RNA-sequencing data from histology stains⁷⁰, or how to reconstruct an automated common coordinate framework from landmarks for a thymus lobe across spatial and cellular protein and RNA profiles⁷⁴.

Successfully achieving an overall unification from the molecular to the histological and anatomical realm will require both experimental and computational efforts. Experimentally, the HCA must collect data deliberately designed for the scale and needs of algorithms, so they can learn appropriate models. The efforts of the BRAIN initiative are especially notable, as the data collected in the mouse brain enabled such models⁷⁵, and a similar effort is now underway for the much larger human brain¹⁸. Computationally, we need algorithms that not only learn shared representations and have generative capacity, but also methods to query those representations for biologically meaningful patterns and insights into tissue biology. This requires both a census of the key spatial features of cells (such as type definitions incorporating morphological features and the cell's environment) and of tissue features (for example, histologies, cell communities and multicellular programs^{72,76–78}). Importantly, algorithms should learn models that both impute missing information and abstract from concrete 3D measurements to generalized coordinates, up to a body-scale common coordinate framework that captures interindividual variability, and links to medical imaging⁶⁷.

The atlas as a genotype-to-phenotype map

The full multiscale HCA must incorporate the genotype and, in this way, serve as a genotype-to-phenotype map. This map would capture key features of human diversity, including those that have been understudied in genomic research to date^{79,80}, where nearly 90% of data collected are from individuals of European ancestry⁸¹. Achieving sufficient statistical power to better encompass human diversity, across ancestry, age, gender and other axes could require at least a tenfold increase in cells/tissues studied than those profiled to date. Given the rapid advances in spatial technologies and their applicability to previously preserved (fixed and/or frozen) material, this expanded data collection may be best combined with spatial profiling efforts. It will also require deep engagement with scientists and communities across the globe, following the framework set by HCA's Equity Working Group, as well as partnering with initiatives focused on common complex disease. Many biobanks (such as UK Biobank, FinnGen and All of US) embrace data beyond human genetics (including peripheral blood mononuclear cell transcriptomics and plasma proteomics and metabolomics), and growth of similar efforts in Asia, Africa and Latin America provides a platform to deliver an atlas of more representative ancestral diversity.

The HCA's Genetic Diversity Biological Network, in partnership with HCA's regional networks in Latin America, Africa, the Middle East and Asia, have spearheaded several efforts to collect a genetically, ancestrally and geographically diverse atlas, beginning with the immune system. For example, the networks' early efforts have helped to identify cell-type-specific regulation of splicing in autoimmune diseases⁸², and variations in breast tissues of women of different ancestries, which may help to explain some of the ancestry-dependent differences in breast cancer⁸³.

Computational analysis is essential for identifying the causal genetic basis of variations in cell and tissue traits. Initially, cell types, states and programs defined by HCA data were combined with human genetics through genes in associated loci⁸⁴ and genome-wide association study (GWAS) summary statistics^{25,85,86}. As atlas efforts advanced, genetic studies directly associated variants to cell-type-specific expression, gene programs and/or cell proportion traits in single-cell data from hundreds of disease tissue samples (for example, post-mortem brain tissue from patients with Alzheimer's disease⁸⁷; peripheral blood mononuclear cells from patients with lupus⁸⁸ or rheumatoid arthritis⁸⁹).

Recent studies, in both human blood cells⁹⁰ and mouse tissue cells, have shown that such analyses can also recover the genetic basis of cell–cell interactions. Thus, as the reference atlases of blood and tissues expand into broad population-based datasets with a genetic component, these will help to translate the atlas into the genetic impact on cell and tissue states in disease.

Generating the atlas as a genotype–phenotype map will necessarily require substantial computational inference, because, unlike germline genomics, we can never aspire to profile all cells of all individuals throughout their entire lifetime. The derived computational map will be crucial to 'fill in the gaps' and—as it is constructed iteratively—to guide data collection efforts to maximize knowledge. By iterating data collection and computational analysis, ML can learn from the data and suggest the next sampling or data needed, to represent a greater swath of humanity. These maps can also be further complemented with perturbational ones collected in disease, in laboratory models and by *in silico* perturbations⁹¹. ML methods that predict missing biological perturbations are steadily improving⁹¹. If successful, this ultimate map would be able to predict the range of cell states or the cell composition of a tissue from an individual's genotype.

The atlas as a temporal map of development

Cells are dynamic throughout the lifespan, but especially during development, a process that has been traditionally hard to study in humans. Mapping human development in 4D is vital to understanding how different cell types and complex tissue architectures are formed in healthy development, and is important for reproductive and regenerative cell therapies.

The Developmental Cell Atlas, which aims to map and characterize cells across all major organs and tissue types at vital stages from fertilization to birth⁹², has already shed substantial light on human development. For example, a multimodal, spatial analysis of prenatal skin⁹³ revealed how skin morphogenesis is regulated by cross-talk between immune and non-immune cells; a study of the human placenta⁹⁴ provides a reference of gene programs that regulate pregnancy and of placenta–uterine interactions^{95,96}; and a reference of healthy cells during brain maturation can shed light on neurodevelopmental disorders⁹⁷. These efforts are complemented by the Organoid Cell Atlas, which reveals insights into developmental biology that cannot be captured in human tissues (obtained from resection, biopsy or post-mortem), and helps to characterize perturbed and diseased states^{13,14}.

Single-cell genomics made it possible to computationally infer biological continuities from discrete snapshot data by learning (pseudo) temporal trajectories from cell profiles⁹⁸. This is one important aspect of a developmental cell atlas but, ultimately, it will also need to include and combine fate maps (relating current cell states to their future potential), lineage maps (tracing series of cell division events) and morphogenesis maps (for cell movement and shape formation), each of which poses key challenges. Many changes are rapid, vary across organ systems, or occur only in very early and inaccessible gestational stages, and all of them proceed in four dimensions, such that a proper developmental atlas must capture morphogenesis. Some of these challenges can be addressed by live measurements (including in human organoid models⁹⁹) from which molecular profiles may be computationally generated⁷¹. In addition to learning representations of individual temporal processes from their measurements, a developmental map could generalize to key patterns (for example, fates and bi/multi-furcations) within and across processes, just like a census recovers cell types and relates 'shared' types across organs.

The Developmental Cell Atlas community is now assembling entire human specimens for comprehensive characterizations, both as single-cell embryonic and fetal atlases^{100–102} and, increasingly, with 2D reconstruction of developmental structures (for example, the embryonic limb⁶⁰). The smaller size of embryos makes it possible to

leapfrog some of the challenges of the adult atlas, while capturing the 3D complexity of multiple tissues at once. It may be that the first full specimen model will be for embryonic stages.

Atlases as foundation models for biology

Various metaphors have been used for the HCA, from ‘Google Maps of the human body’ to a ‘periodic table of our cells’. The Google Maps analogy captures its multiscale goal and dynamic nature, whereby the navigation function is informed by the data of multilayered maps (down to street map level of granularity)¹; and the periodic table analogy captures the aspiration to form a theory, from which as-yet unknown entities and their properties can be predicted before they are observed.

The rapid computational advances in learning foundation models, which are large-scale, pretrained machine learning models with general-purpose capabilities across a wide range of downstream tasks, have opened the way to learn models of the atlas that serve each of the four functions above, while also providing a representation that captures more than a compact description of its measured training data, including generating predictions beyond our observations. The promise of ‘filling in the blanks’ is crucial, as HCA never intended nor can be expected to measure everything, just enough to be able to generalize the rest.

The current data from the HCA have already provided the unique large-scale data needed to train foundation models. The first such foundation models, including GeneFormer³³, scGPT³⁴, scFoundation¹⁰³ and the Universal Cell Embedding¹⁰⁴, all relied heavily on HCA data, and focused on common well defined tasks, such as integration of new data, cell annotations, identification of marker gene signatures and prediction of perturbation outcomes or of perturbations that would yield desired cell states. Recently, methods like SCimilarity²² have focused on learning foundation models that perform well also in newly-defined tasks, such as querying for entire cell profiles across a full atlas.

We expect many other tasks to be tackled using foundation models. Notable opportunities include the prediction of new cell types that were not observed experimentally (which can at first be demonstrated by withholding entire sublineages at training), and generating cell states from different genetic profiles (natural ones, in the context of a genotype–phenotype map; or engineered ones, including for regenerative cell therapy). While these tasks have not been accomplished, and may be highly challenging, achieving them would indicate that the model has captured salient biological features well beyond its training. Such predictions would rely on the ability to test predictions experimentally. Foundation models would also be crucially needed for integrating across scales (for the cell atlas as a multimodal map) and for integrating across the body (for the ultimate cell census). Moreover, a foundation model can extend beyond the realm of a single species, either to allow queries from another species, or to be learned from data from two or more species as part of an ‘AI virtual cell’¹⁰⁵, including model organisms such as fly and mouse, as has already been explored in Universal Cell Embeddings¹⁰⁴.

These goals for foundation models may seem ambitious to achieve, but some efforts have shown early successes, such as for cross-modality predictions⁶⁹. Methods to link across multimodal data will be critical for predicting the nonlinear transformation between layers, and to infer their causal relationships. However, other use cases may be more difficult or prove to be impossible to realize, such as predicting entirely new cell types. Nevertheless, attempting to follow them, and understanding the power and limits of a model will teach us about the dependencies between cells. Notably, studies comparing the performance of the first generation of foundation models to simpler established approaches showed that, at times, the latter may outperform the former in specific tasks^{106–108}. However, the concept of foundation models is much broader than any initial instance, in their focus on training on large-scale, diverse

datasets (for example, in contrast to a single tissue), adaptability to a wide variety of downstream tasks (for example, in contrast to a single purpose model), scalability, high transferability and multimodality. Foundation models, as they are today, are not a panacea, and the first generation of models relied on architectures as established for other domains, but exploring additional architectures could help overcome some of their current challenges, as well as better understand any inherent limitations. To maximize the predictive power of foundation (and other) models, we will need targeted data collection efforts, in addition to the harmonized and curated cell atlases described above. In particular, the community will need to iterate between wet laboratory data generation and model output to ensure that the predictions of the models reflect the biology of the system.

As foundation models advance, we will need to grow our expectations of the questions and queries to ask of an atlas, and will need to understand the inner workings of the representation learned by the model. For example, in a temporal/developmental context, we will want to identify shared patterns and processes that govern transitions in multiple biological scenarios (for example, immediate early gene activation, the cell cycle and T cell polarization are processes that happen through one principle in multiple cell types). Learning such shared patterns in cell responses or tissue organization will provide us with new words and sentences for the language of biology.

Transforming medicine with cell atlases

The laboratory methods, reference data and algorithms developed by the HCA community have already become an integral part of biomedical research from fundamental biology of the healthy body to translation¹⁰⁹ through disease cell atlases, experimental medicine¹¹⁰ and clinical trials that use atlas technologies for biomarker analysis. For example, a comprehensive reference of circulating immune cells across acute and chronic inflammatory conditions¹¹¹ revealed disease-specific inflammatory cell states and T cell dynamics, which can help us to understand disease mechanisms, expand indications and identify new universal diagnostics. Moreover, tissue analysis along the gastrointestinal tract characterized gastric mucous cell metaplasia in intestinal inflammation (celiac disease and inflammatory bowel disease) and in colorectal cancer¹⁵. The HCA is also the starting point for deep endophenotyping of diseases into their constituent subtypes, which will improve patient stratification and diagnosis, including through creation of enhanced assays for tissue pathology. For example, cell atlas data of the nasal epithelium revealed distinguishing features in individuals more protected from SARS-CoV-2 infection¹¹⁰.

Cell atlases are now moving from descriptive cell labels (annotations) in a cell census to 3D and 4D (temporal) maps, as well as causal genotype-to-phenotype maps. Combining learnings from atlases, including those of model organisms^{9,20}, with human genetic data, longitudinal donor and patient samples as well as high-throughput perturbation screens will further unlock mechanisms of human health and disease, to increase the predictive power of our models.

Despite decades of medical research advances, there is still an enormous need to understand how gene activities generate cellular diversity in the body, and how disruptions in these processes cause disease. With the past decade’s ‘resolution revolution’ in genomics, combined with findings from the laboratory, we can now build molecular maps of individual cells at an astonishing scale, and connect them to their function in tissues, but there is still work to do to use atlases to reveal complex mechanistic and functional aspects of biology. This work will need to include experimental perturbations to probe, test and assess the impact of particular genes and variants, leveraging multiple available datasets to contribute towards building a complementary ‘perturbation cell atlas’⁹¹. With AI to help interpret these data and integrate them with new spatial profiling and imaging, we are now in the midst of this extraordinary opportunity to build comprehensive 4D maps

of the human body. The healthy reference maps from the HCA—as a cell census, a multiscale 3D map, a genotype-to-phenotype map, a developmental map and, ultimately, a foundation model—will be the underpinning framework that provides the gold-standard baseline for understanding, monitoring, diagnosis and treatment in health and disease.

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Acknowledgements This publication is part of the HCA (<https://www.humancellatlas.org/publications>). S.W., L.R. and J.R. are members of the HCA Executive Office, which has been core funded by the Chan Zuckerberg Initiative, Helmsley Charitable Trust, the Klarman Family Foundation and Wellcome.

Author contributions A.R. and S.A.T. conceived the Perspective. J.E.R., S.W. and A.R. wrote the manuscript. A.R., J.E.R., L.R., J.R. and S.A.T. developed and edited the manuscript, and A.R., L.R., J.E.R. and S.A.T. revised the manuscript after review. A.H. created the figure.

Competing interests J.E.R., A.H. and A.R. are employees of Genentech, a member of the Roche Group and have equity in Roche. A.R. is a co-founder and equity holder of Celsius Therapeutics, an equity holder in Immunitas and, until 31 July 2020, was an scientific advisory board member of Thermo Fisher Scientific, Syros Pharmaceuticals, Neogene Therapeutics and Asimov. A.R. is a named inventor on multiple filed patents related to single-cell and spatial genomics. In the past three years, S.A.T. has consulted for or been a member of scientific advisory boards at Qiagen, Sanofi, GlaxoSmithKline and ForeSite Labs. S.A.T. is a consultant and equity holder for TransitionBio and EnsoCell. From 8 January 2024, S.A.T. is a part-time employee of GlaxoSmithKline.

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Peer review information *Nature* thanks Rafael Kramann, Hongkui Zeng and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.
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